Testing Report

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1. Test Setup

- Test Set Size: 1,179 samples (stratified split)

- Class Distribution: Drone (393), Normal (393), Jamming (393)

- Evaluation Metrics: Accuracy, Precision, Recall, F1-Score, Confusion Matrix

- Hardware: Kaggle GPU (Tesla T4 x2)

2. Overall Results

- Accuracy: 93.47%

- Total Correct: 1,102 / 1,179

- Total Errors: 77 (6.53%)

3. Per-Class Performance

	Precision	Recall	F1-Score	Support
Drone	1.00	0.80	0.89	393
Normal	0.97	1.00	0.98	393
Jamming	0.86	1.00	0.92	393

Macro Avg: Precision=0.94, Recall=0.93, F1=0.93

4. Confusion Matrix

Predicted

True	Drone	Normal	Jamming
Drone	316	13	64
Normal	0	393	0
Jamming	0	0	393

5. Error Breakdown

- Drone 64 errors (misclassified as Jamming)

- Normal 0 errors

- Jamming 0 errors

6. Conclusion

The model achieves excellent performance on Normal and Jamming classes (100% recall). Drone class requires improvement, primarily due to signal diversity. The model is ready for integration with SDR hardware and real-time deployment. Recommended next steps: fine-tuning on drone-specific data or collecting more diverse drone samples.